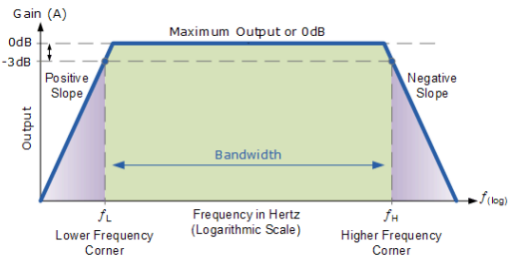
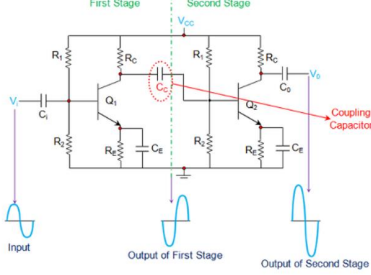
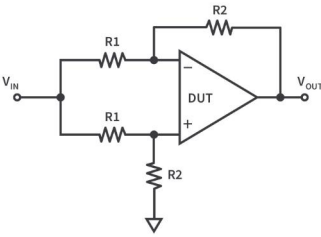
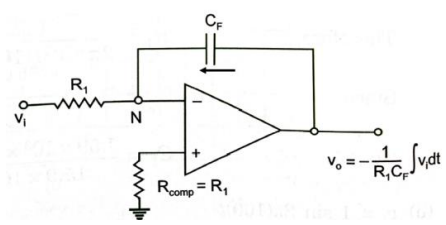
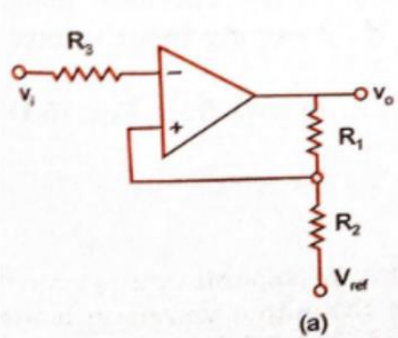
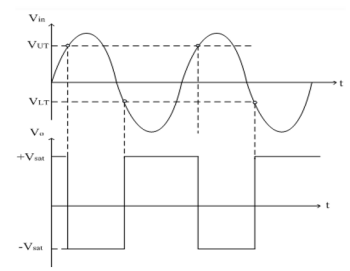
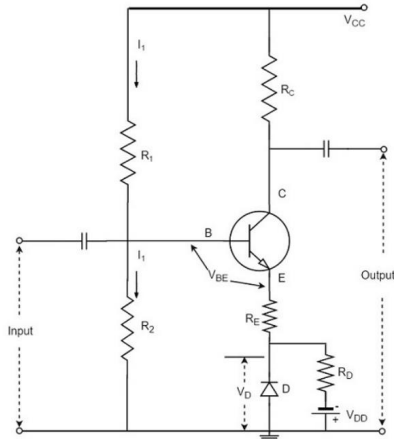
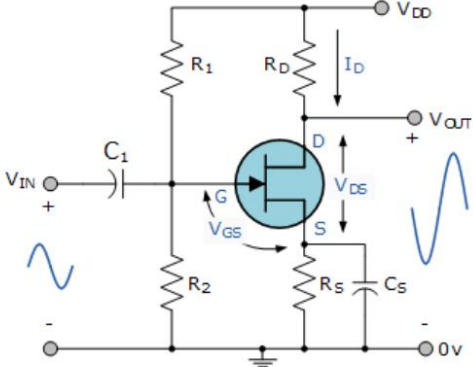


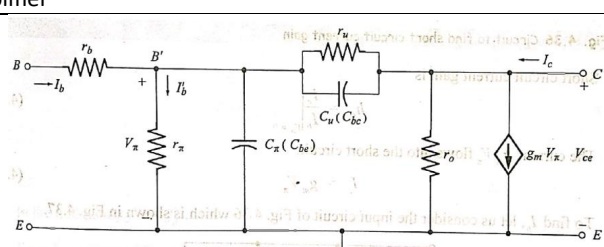
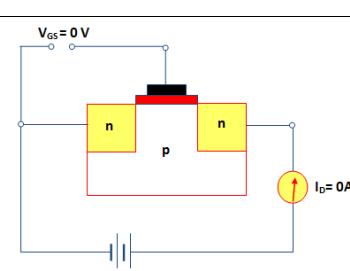
Sahrdaya College of Engineering and Technology, Kodakara	
Answer Key	
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY	
THIRD SEMESTER B.TECH DEGREE EXAMINATION, DECEMBER 2020	
Course Code: EET205	
Course Name: ANALOG ELECTRONICS	
PART A	
	<i>Answer all questions. Each question carries 3 marks</i>
Q1	Explain why a fixed bias circuit is not widely used in amplifiers, despite its simplicity
ans	To ensure absolute stability of the amplifier, a stability factor of less than 25 is preferred, and so small-signal transistors have large stability factors. Usage: Due to the above inherent drawbacks , fixed bias is rarely used in linear circuits (i.e., those circuits which use the transistor as a current source)
Q2	Determine the following parameter for the collector to base bias circuit of transistor amplifier circuit i) I_B ii) I_C .
ans	$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)}$ $I_C = \beta I_B$
Q3	Define the following JFET parameters: (a) Transconductance (b) Dynamic output resistance
ans	Transconductance is the ratio of change in drain current (δI_D) to change in the gate to source voltage (δV_{GS}) at a constant drain to source voltage ($V_{DS} = \text{Constant}$). $g_m = \frac{\delta I_D}{\delta V_{GS}} \text{ at constant } V_{DS}$ This is the ratio of change of drain to source voltage (δV_{DS}) to the change of drain current (δI_D) at a constant gate to source voltage ($V_{GS} = \text{Constant}$). The ratio is denoted as r_d. $r_d = \frac{\delta V_{DS}}{\delta I_D} \text{ at constant } V_{GS}$
Q4	Draw the frequency response of CE amplifier and explain the concept of bandwidth
ans	
Q5	Draw the circuit of two stage RC coupled amplifier and briefly explain loading effect

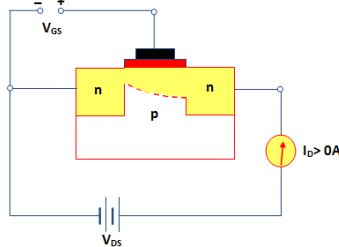
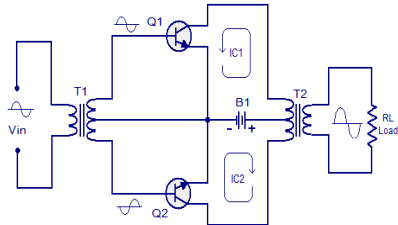
ans	 Overall gain is less than the product of gains of individual stages due to loading effect (load resistance of first stage reduced due to shunting effect of second stage input R_i)
Q6	State and explain Barkhausen's criterion of oscillation
ans	According to Barkhausen criterion an amplifier will work as an oscillator if and only if its total phase difference should be zero or multiple of 2π and $A\beta = 1$ i.e magnitude of feedback factor should be unity. The gain of the positive feedback amplifier is given by $A_f = \frac{A}{(1 - A\beta)}$
Q7	Compare the characteristics of ideal Op-Amps and practical Op-Amps
ans	
Q8	Explain the concept of common mode rejection with suitable diagram.
ans	$\Delta V_{OUT} = \frac{\Delta V_{IN}}{CMRR} \left(1 + \frac{R_2}{R_1} \right)$ Resistors must match within 1 ppm (0.0001%) to measure CMRR > 100dB $CMRR = \frac{\Delta V_{IN} \left(1 + \frac{R_2}{R_1} \right)}{\Delta V_{OUT}}$  $CMRR = \frac{A_{v(d)}}{A_{cm}}$
Q9	Draw the circuit diagram of an ideal integrator and derive the expression for output voltage.

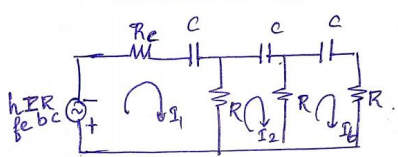
ans	 Nodal equation at node A $\frac{v_i}{R_1} + C_F \frac{dv_o}{dt} = 0$ or, $\frac{dv_o}{dt} = -\frac{1}{R_1 C_F} v_i$ Integrating both sides, we get $\int_0^t dv_o = -\frac{1}{R_1 C_F} \int_0^t v_i dt$
Q10	Explain voltage level detector circuit with neat diagram. Draw the output waveform considering that the input is $V_{in}=10\sin\omega t$ and $V_{ref}=4V$
ans	 (a) 
PART B	
<i>Answer any one full question from each module. Each question carries 14 marks</i>	
Module 1	
Q11a	Determine the voltage gain, current gain, input impedance and output impedance of the common emitter amplifier circuit driving a load of 5kΩ. It is supplied by a signal source of internal resistance 10kΩ. The hybrid parameters of the transistors are $h_{ie}=1100\ \Omega$, $h_{re}=2.5\times 10^{-4}$, $h_{fe}=50$, $h_{oe}=25\ \mu\text{S}$

ans	$A_i = \frac{-h_f}{1 + h_o R_L}$ $R_i = h_i + h_r A_i R_L$ $Z_o = \frac{1}{h_{oe} - \frac{h_{fe} h_{re}}{(R_s + h_{ie})}}$
Q11b	Explain any one compensation technique used for reducing the drift of operating point
ans	 In a Silicon transistor, the changes in the value of V_{BE} results in the changes in I_C. A diode can be employed in the emitter circuit in order to compensate the variations in V_{BE} or I_{CO}. As the diode and transistor used are of same material, the voltage V_D across the diode has same temperature coefficient as V_{BE} of the transistor. The diode D is forward biased by the source V_{DD} and the resistor R_D. The variation in V_{BE} with temperature is same as the variation in V_D with temperature, hence the quantity $(V_{BE} - V_D)$ remains constant. So the current I_C remains constant in spite of the variation in V_{BE}.
Q12a	Determine the following parameter for the fixed bias configuration of transistor amplifier. i) I_C ii) V_{CC} iii) R_B iv) β Given $R_c = 2.5k\Omega$, $I_B = 20\mu A$, $I_E = 3mA$, $V_{CE} = 6.5V$
ans	
Q12b	List the factor affecting the stability of operating point of a transistor. Derive the expression for stability of Voltage divider bias
ans	- The main factor that affect the operating point is the temperature. The operating point shifts due to change in temperature. - As temperature increases, the values of I_C, β, V_{BE} gets affected. I_{CBO} gets doubled (for every 10° rise) V_{BE} decreases by 2.5mv (for every 1° rise) - So the main problem which affects the operating point is temperature. Hence operating point should be made independent of the temperature so as to achieve stability. To achieve this, biasing circuits are introduced.

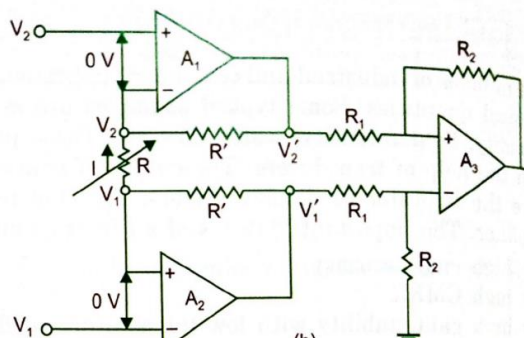
		$\frac{dI_B}{dI_C} = \frac{-R_E}{R_E + R_T}$ Stability factor, $S = \frac{1 + \beta}{1 - \beta \cdot \frac{dI_B}{dI_C}}$ $S = \frac{1 + \beta}{1 - \beta \left(\frac{-R_E}{R_E + R_T} \right)}$ $S = \frac{1 + \beta}{1 + \frac{\beta R_E}{R_E + R_T}} = \frac{(1 + \beta)(R_E + R_T)}{(R_E + R_T) + \beta R_E}$ ∴ Neg & Dis with R_E $S = (1 + \beta) \cdot \frac{1 + \frac{R_T}{R_E}}{1 + \beta + \frac{R_T}{R_E}}$
Module 2		
Q13a	Draw and explain the working of common drain FET amplifier with voltage divider bias arrangement using small signal AC equivalent circuit. Also derive the expression for voltage gain, input impedance and output impedance.	
ans	 ① <u>Input Impedance</u> :- $R_{in} = R_G = R_1 \parallel R_2$ ② <u>Output Impedance</u> :- $Z_{out} = \frac{1}{g_m} \parallel r_d \parallel R_S$ Usually $r_d \gg \frac{1}{g_m}$ $\therefore Z_{out} = \frac{1}{g_m} \parallel R_S$ ③ <u>Voltage gain</u> :- $V_{out} = I_D [R_S \parallel r_d \parallel R_L]$ $= g_m V_{GS} [R_S \parallel r_d \parallel R_L]$	

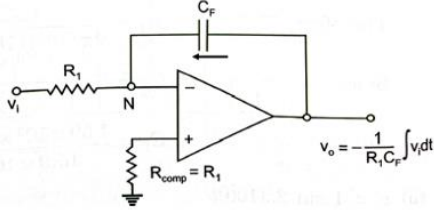
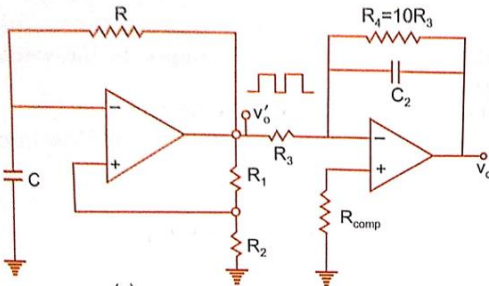
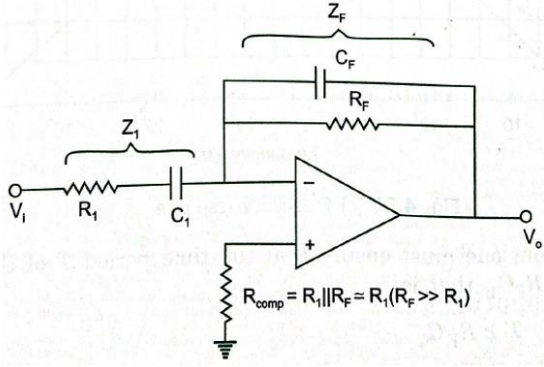
Q13b	Mention any four applications of FET.
ans	• Low Noise Amplifier. • Buffer Amplifier. • Cascode Amplifier. • Analog Switch. • Chopper. • Multiplexer. • Current Limiter.
Q14a	With suitable circuit diagrams, explain the effect of internal capacitance during high frequency operation of CE amplifier
ans	 • In above model the capacitor between base and collector is denoted as C_{μ}. It is also called as collector junction capacitance. • Its value is small in the range of 0.5 to 5pF. Its value varies with the width of the junction depletion region. • The capacitor between base and emitter is denoted as C_{π}. • C_{π} is the sum of two parts: C_{de} the emitter diffusion capacitance [that is proportional to the dc bias current] and C_{je} an emitter junction depletion layer capacitance (that is dependent on V_{BE}). i.e. $C_{\pi} = C_{de} + C_{je}$ • Since the diffusion capacitance is larger than the depletion layer capacitance, C_{π} is approximately equal to diffusion capacitance (C_{de}). • They are alternatively represented by C_{be} and C_{bc} respectively. • Both these makes h_{fe} to vary with frequency in high frequency range • The resistors are the resistances between the indicated terminals of the transistor when the transistor is in the active region. • The resistance is identical to introduced in the common – emitter r_e model. The resistance r_u represents the feedback from collector to base. It is very large typically in the mega ohm range. • r_o is the output resistance which appears between the collector and emitter terminals
Q14b	Explain the operation of E-type MOSFET
ans	

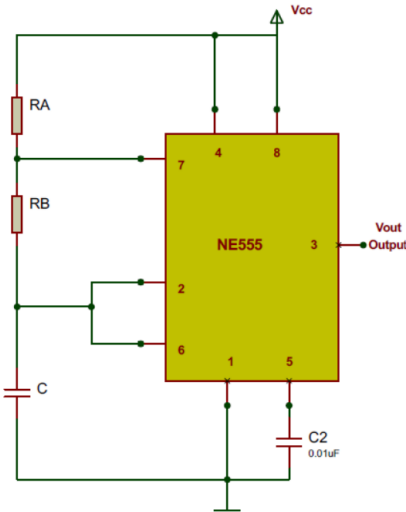
		(i) When $V_{GS} = 0V$, as shown in fig.9 (i), there is no channel connecting source and drain. - The p-substrate has only a few thermally produced free electrons(minority carriers) so that drain current is almost zero. For this reason, E-MOSFET is normally OFF when $V_{GS} = 0V$.  (ii) When V_{GS} is positive, i.e gate is made positive as shown in fig.9(ii), it attracts free electrons into the p region. The free electrons combine with the holes next to the SiO₂ layer. - If V_{GS} is positive enough, all the holes touching the SiO₂ layer are filled and free electrons begin to flow from the source to drain. The effect is same as creating a thin layer of n-type material i.e. inducing a thin n-layer adjacent to the SiO₂ layer (iii) When V_{GS} is less than $V_{GS(th)}$, there is no induced channel and the drain current I_D is zero.
Module 3		
Q15a	Describe the operation of class B push pull power amplifier and determine its maximum power conversion efficiency.	
ans		 - When no signal is applied at the input, the transistors Q1 and Q2 are in cut off condition and hence no collector currents flow. As no current is drawn from V_{CC}, no power is wasted. - When input signal is given, it is applied to the input transformer T1 which splits the signal into two signals that are 180° out of phase with each other. - These two signals are given to the two identical transistors Q1 and Q2. - For the positive half cycle, the base of the transistor Q1 becomes positive and collector current flows. At the same time, the transistor Q2 has negative half cycle, which throws the transistor Q2 into cutoff condition and hence no collector current flows $\therefore I_{dc} = \frac{I_{C(sat)}}{\pi}$ $\therefore P_{dc} = \frac{V_{CC} I_{C(sat)}}{\pi}$ $\therefore \text{Max. collector } \eta = \frac{P_{o(max)}}{P_{dc}} = \frac{0.25 V_{CC} I_{C(sat)}}{(V_{CC} I_{C(sat)})/\pi} \times 100 = 0.25\pi \times 100 = 78.5\%$

Q15b	List any two advantages, disadvantages and applications of Transformer coupled amplifier.										
ans	- The efficiency of the operation is higher than directly coupled amplifier. - The impedance matching is required for maximum power transfer is possible - Due to the inductance of the transformer winding, the frequency response is poor compared to series fed amplifier. - The circuit is bulky, heavy and costly due to the transformer <u>Application</u> - Mostly used for impedance matching purposes. - Used for Power amplification. - Used in applications where maximum power transfer is needed										
Q16a	Compare positive and negative feedback in amplifiers										
ans	<table> <tr> <th>Positive Feedback</th><th>Negative Feedback</th></tr> <tr> <td>0° or 360°</td><td>180°</td></tr> <tr> <td>Non-inverting terminal of op-amp</td><td>Inverting Terminal of op-amp</td></tr> <tr> <td>Low</td><td>High</td></tr> <tr> <td>In oscillators.</td><td>In amplifiers.</td></tr> </table>	Positive Feedback	Negative Feedback	0° or 360°	180°	Non-inverting terminal of op-amp	Inverting Terminal of op-amp	Low	High	In oscillators.	In amplifiers.
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Low	High										
In oscillators.	In amplifiers.										
Q16b	With a neat circuit diagram, explain RC phase shift oscillator using BJT. Derive an expression for frequency of oscillation										
ans	<u>RC-Phase shift Oscillator</u> <u>frequency of oscillation</u>  $\begin{vmatrix} (R+R_c-jX_c) & -R & h_{fe}R_c \\ (-R) & (2R-jX_c) & (-R) \\ 0 & -R & (2R-jX_c) \end{vmatrix} = 0$ $(R+R_c-jX_c)[(2R-jX_c)^2 - R^2] + R[-R(2R-jX_c)] + h_{fe}R_c R^2 = 0$ Separate Real & Imaginary parts $\Rightarrow R^3 - 5RX_c^2 - X_c^2 R_c + 3R^2 R_c + h_{fe}R_c R^2 - 6jR^2 X_c - j4RR_c X_c + jX_c^3 = 0$ $\Rightarrow R^3 - 5RX_c^2 - X_c^2 R_c + (3+h_{fe})R^2 R_c - j6R^2 X_c - j4RR_c X_c + jX_c^3 = 0$										

	$6R^2X_C + 4RR_CX_C - X_C^3 = 0$ $X_C = \sqrt{6R^2 + 4RR_C}$ $2\pi fC = \frac{1}{\sqrt{6R^2 + 4RR_C}}$ $f = \frac{1}{2\pi RC\sqrt{6 + 4\frac{R_C}{R}}}$ $f = \frac{1}{2\pi RC\sqrt{10}}$ where $K = \frac{R_C}{R}$																																																												
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Q17a	Compare the characteristics of ideal Op-Amps and practical Op-Amps.																																																												
ans	<u>Comparison of Op-Amps</u> <table><tr><th>Parameter</th><th>Ideal</th><th>Practical</th><th>IC 741</th></tr><tr><td>Voltage Gain</td><td>Infinite</td><td>Order of 10^5</td><td>2×10^5 for LF</td></tr><tr><td>Input Resistance</td><td>Infinite</td><td>Order of $M\Omega$</td><td>$2M\Omega$</td></tr><tr><td>Output Resistance</td><td>Zero</td><td>Tens of Ohms</td><td>75Ω</td></tr><tr><td>Bandwidth</td><td>Infinite</td><td>Restricted by Slew Rate</td><td>1MHz (Unity B.W)</td></tr><tr><td>CMRR</td><td>Infinite</td><td></td><td>90dB</td></tr><tr><td>Slew Rate</td><td>Infinite</td><td></td><td>$0.5 V/\mu s$</td></tr><tr><td></td><td>Zero output voltage when there is no input voltage</td><td></td><td></td></tr><tr><td>Power Supply</td><td></td><td></td><td>5V -18V</td></tr><tr><td>Max Output Current</td><td></td><td></td><td>20mA</td></tr><tr><td>Recommended Output Load</td><td></td><td></td><td>$> 2K\Omega$</td></tr><tr><td>Input Offset Voltage</td><td></td><td></td><td>2mv - 6mV</td></tr><tr><td>Input Offset Current</td><td></td><td></td><td>200nA - 6nA</td></tr><tr><td>Supply Current</td><td></td><td></td><td>2.8mA</td></tr><tr><td>Power Consumption</td><td></td><td></td><td>85mV</td></tr></table>	Parameter	Ideal	Practical	IC 741	Voltage Gain	Infinite	Order of 10^5	2×10^5 for LF	Input Resistance	Infinite	Order of $M\Omega$	$2M\Omega$	Output Resistance	Zero	Tens of Ohms	75Ω	Bandwidth	Infinite	Restricted by Slew Rate	1MHz (Unity B.W)	CMRR	Infinite		90dB	Slew Rate	Infinite		$0.5 V/\mu s$		Zero output voltage when there is no input voltage			Power Supply			5V -18V	Max Output Current			20mA	Recommended Output Load			$> 2K\Omega$	Input Offset Voltage			2mv - 6mV	Input Offset Current			200nA - 6nA	Supply Current			2.8mA	Power Consumption			85mV
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Q17b	Draw the circuit of an inverting amplifier and obtain the expression for its closed loop gain. Also design an inverting amplifier with gain 12.																																																												
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		$I_{in} = I_f + i_b$ $i_b = 0$ $I_{in} = I_f$ $\frac{V_{in} - V_2}{R_1} = \frac{V_2 - V_{out}}{R_f}$ $V_1 = V_2 = 0V$ $\frac{V_{in}}{R_1} = \frac{-V_{out}}{R_f}$ $\frac{V_{out}}{V_{in}} = -\frac{R_f}{R_1}$ $V_{out} = \left(-\frac{R_f}{R_1}\right)V_{in}$
Q18a	Design an op-amp circuit to obtain an output of $V_o = -[3V_1 + 2V_2 + 0.5V_3 + 5V_4]$. where V_1, V_2, V_3 and V_4 are the inputs to op-amp	
ans		$V_o = -\left(\frac{R_f}{R_1}V_1 + \frac{R_f}{R_2}V_2 + \frac{R_f}{R_3}V_3\right)$
Q18b	What are the features of an instrumentation amplifier using op-amp? Derive the expression for output voltage.	
ans		 (b) - Since, no current flows into op amp, the current (I) flowing (upwards) in R is $I = (V_1 - V_2)/R$ and passes through the resistor R'. i.e., $V_1' = IR' + V_1 = \frac{R'}{R}(V_1 - V_2) + V_1$ and $V_2' = -IR' + V_2 = -\frac{R'}{R}(V_1 - V_2) + V_2$ - Substituting the values of V_1' and V_2' in Eqn.(1), we obtain $V_o = \frac{R_2}{R_1} \left\{ \left(\frac{R'}{R}(V_1 - V_2) + V_1 \right) + \left(\frac{R'}{R}(V_1 - V_2) + V_2 \right) \right\}$ $V_o = \frac{R_2}{R_1} \left(\frac{2R'}{R}(V_1 - V_2) + (V_1 + V_2) \right)$ $V_o = \frac{R_2}{R_1} \left(1 + \frac{2R'}{R} \right) (V_1 - V_2)$
Module 5		
Q19a	Explain the working of an ideal integrator using op-amp and derive the expression for output voltage. Draw the output waveform of the integrator when input is square wave.	

ans	 Nodal equation at node A $\frac{v_i}{R_1} + C_F \frac{dv_o}{dt} = 0$ or, $\frac{dv_o}{dt} = -\frac{1}{R_1 C_F} v_i$ Integrating both sides, we get $\int_0^t dv_o = -\frac{1}{R_1 C_F} \int_0^t v_i dt$ 
Q19b	What are the disadvantages of ideal differentiator and what are the modifications done in practical differentiator to overcome this?
ans	Limitations of an Ideal Differentiator At high frequencies-This simple differentiator circuit becomes unstable and starts to oscillate. The circuit becomes sensitive to noise, that is, when amplified, noise dominates the input/ message signal. 

	From Eq. (4.73), it is evident that the gain increases at +20 dB/decade for frequency $f < f_b$ and decreases at -20 dB/decade for $f > f_b$ as shown by dashed lines in Fig. 4.21 (c). This 40 dB/decade change in gain is caused by $R_1 C_1$ and $R_F C_F$ factors. For the basic differentiator of Fig. 4.21 (a), the frequency response would have increased continuously at the rate of +20 dB/decade even beyond f_b causing stability problem at high frequency. Thus the gain at high frequency is reduced significantly, thereby avoiding the high frequency noise and stability problems. The value of f_b should be selected such that,
Q20a	With the help of internal circuit diagram explain the working of an astable multivibrator. Derive the expression for frequency of oscillation
ans	 ➤ Connect external timing capacitor between trigger point (pin 2) and Ground. ➤ Split external timing resistor R into R1 & R2, and connect their junction to discharge terminal (pin 7). ➤ Remove trigger input, monostable is converted to Astable multivibrator. ➤ This circuit has no stable state. The circuit changes its state alternately. Hence the operation is also called free running oscillator. • Resistive voltage divider (equal resistors) sets threshold voltages for comparators $V_1 = V_{TH} = 2/3 V_{CC}$ $V_2 = V_{TL} = 1/3 V_{CC}$ • Two Voltage Comparators - For A1, if $V_+ > V_{TH}$ then R = HIGH - For A2, if $V_- < V_{TL}$ then S = HIGH • RS FF • - If S = HIGH, then FF is SET, = LOW, Q1 OFF, $V_{OUT} = \text{HIGH}$ - If R = HIGH, then FF is RESET, = HIGH, Q1 ON, $V_{OUT} = \text{LOW}$ • Transistor Q1 is used as a Switch <i>Total Time period</i> $T = t_1 + t_2 = 0.693(R_1 + 2R_2)C \text{ sec}$ <i>frequency</i> $f = \frac{1.45}{(R_1 + 2R_2)C} \text{ Hz}$
Q20b	Design an astable multivibrator using 555 timer IC to generate an output signal with frequency 2kHz and 60% duty cycle
ans	

		$f = 1/(0.693C(R_A + R_B))$ $\text{i.e. } 103 = 1/(0.693C(R_A + R_B))$ Assume $C = 0.1\mu_F$ $\text{Thus } R_A + R_B = 14400 \dots (4)$ From eq (3) & (4) we get, $R_A = 5.76K\Omega$ $R_B = 8.64K\Omega$
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